

Tatevik Mkrtchyan

PhD Student in Astrophysics, Universidad Diego Portales, Santiago, Chile
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RESEARCH INTERESTS

Active Galactic Nuclei, High-redshift quasars, Galaxy Evolution and Formation, Supermassive Black Holes, Photometric Surveys

EDUCATION

PhD Student in Astrophysics August 2022 — April 2027
Instituto de Estudios Astrofísicos, Universidad Diego Portales Santiago, Chile
Thesis Title: Characterizing Active Galactic Nuclei across Cosmic Time

Master of Science in Astrophysics September 2019 — May 2021
Faculty of Physics, Yerevan State University Yerevan, Armenia
Thesis Title: The Study of High-Energy Gamma-Ray Emission from Galaxy Clusters

Bachelor of Science in Physics September 2014 — June 2019
Faculty of Physics, Yerevan State University Yerevan, Armenia
Thesis Title: High-Energy Gamma-Ray Radiation Processes of Blazars

ACADEMIC EXPERIENCE

Visiting Researcher November 2025 — December 2026
Gemini North Observatory Hilo, Hawaii

- Worked with Dr. Emanuele Farina on the study of Lyman-alpha halos around high-redshift radio-loud quasars.

Senior Lab Assistant April 2021 — September 2022
Byurakan Astrophysical Observatory Byurakan, Armenia

- Worked on Galaxy Evolution models using Principal Component Analysis.

Junior Researcher Intern December 2016 — March 2018
International Center for Relativistic Astrophysics Yerevan, Armenia

- Made models of high-energy gamma-ray radiation processes of blazars using Wolfram Mathematica.

PUBLICATIONS

Refereed publications

1. 4MOST ChANGES: Catalog of high-redshift quasar candidates ($4.5 < z < 7$) selected with SED fitting
T. Mkrtchyan, C. Mazzucchelli, R. J. Assef, M. J. Temple, et al. 2026, A&A, 710, A170
2. VAR-PZ: Constraining the photometric redshifts of quasars using variability
S. Satheesh-Sheeba, R. J. Assef, T. Anguita, P. Sánchez-Sáez, . . . , **T. Mkrtchyan**, et al. 2026, A&A, 708, A121
3. Discovery and characterization of 25 new quasars at $4.6 < z < 6.9$ from wide-field multi-band surveys
S. Belladitta, E. Bañados, Z.-L. Xie, R. Decarli, . . . , **T. Mkrtchyan**, et al. 2025, A&A, 699, A335

Non-refereed publications

1. Photometric redshifts for active galactic nuclei with LePHARE for the Vera C. Rubin Observatory
R. Shirley, M. Salvato, J. Cohen-Tanugi, O. Ilbert, . . . , **T. Mkrtchyan**, et al. 2026, arXiv:2608.30494
2. VAR-PZnn: A machine-learning framework for AGN photometric redshifts using color and variability-based features
S. Satheesh-Sheeba, P. Sánchez-Sáez, R. J. Assef, T. Anguita, . . . , **T. Mkrtchyan**, et al. 2026, arXiv:2607.16434

Books and educational publications

1. Inclusive Teaching Methods Handbook
Published by the Republican Pedagogical-Psychological Center of the Republic of Armenia, 2023

Full record: ADS Publication Profile

OBSERVING PROPOSALS (PI/Co-I)

PI – Gemini/GMOS-S, GS-2026B-Q-405 <i>Radio Needles at Cosmic Dawn: Spectroscopic Confirmation of Southern Radio-Loud Quasars</i>	38 hours
Co-I – ALMA, 2026.1.01357.S <i>An ALMA View of a Potential Protocluster around a Radio-Loud Quasar at Cosmic Dawn</i>	18 hours
Co-I – VLT/MUSE, 116.28SJ <i>A MUSE Snapshot Survey of the First Radio-Loud Quasars</i>	50 hours
Co-I – VLT/MUSE, 114.27A5 <i>A MUSE Snapshot Survey of the First Radio-Loud Quasars</i>	42 hours
Co-I – NTT/EFOSC2, 114.27MF <i>The Ultimate EFOSC2 Legacy Survey for Bright High-z Quasars</i>	48.8 hours

OBSERVATIONS

Observing run with Baade Magellan Telescope/IMACS <i>Las Campanas Observatory, 1 night</i>	La Serena, Chile September 2026
Observing run with NTT/EFOSC-2 <i>La Silla Observatory, 4 nights</i>	La Serena, Chile January 2024
Observing run with Baade Magellan Telescope/FIRE <i>Las Campanas Observatory, 2 nights</i>	La Serena, Chile August 2023
Observing run with Baade Magellan Telescope/FIRE <i>Las Campanas Observatory, 1 night</i>	La Serena, Chile May 2023
Observing run with Baade Magellan Telescope/FIRE <i>Las Campanas Observatory, 2 nights</i>	La Serena, Chile April 2023

CONFERENCES AND PRESENTATIONSConferences

European Astronomical Society Annual Meeting 2026 European Astronomical Society (<i>Contributed Talk</i>)	Jul 2026 Lausanne, Switzerland
The Baryon Cycle from Reionization to Cosmic Noon South American Center for Astronomy, Chinese Academy of Sciences (<i>Contributed Talk</i>)	Dec 2025 Puerto Varas, Chile
Highly Accreting Supermassive Black Holes Across all Cosmic Times European Southern Observatory (<i>Contributed Talk</i>)	Dec 2025 Santiago, Chile
SOCHIAS XX Annual Meeting SOCHIAS, Universidad Austral de Chile (<i>Contributed Talk & Poster; Best AGN Student Talk</i>)	Oct 2025 Puerto Montt, Chile
Extreme Galaxies in their Extreme Environments at Extremely Early Epochs The National Institute for Astrophysics (<i>Contributed Poster</i>)	May 2024 Reykjavik, Iceland
The Santiago AGN Workshop: A Holistic View of Active Galaxies European Southern Observatory (<i>Contributed Talk</i>)	Nov 2023 Santiago, Chile
New Era of AGN Science with the Vera C. Rubin LSST LSST AGN Science Collaboration, National Radio Astronomy Observatory (<i>Contributed Talk</i>)	Jul 2023 Charlottesville, USA

Seminars

Weekly Seminar The Alikhanyan National Science Laboratory, Center for Cosmology and Astrophysics	Jun 2026 Yerevan, Armenia
Weekly Seminar European Southern Observatory, Extragalactic Group	Feb 2026 Santiago, Chile
Astronomy Tuesday Talk University of California, Los Angeles (UCLA)	Nov 2025 Los Angeles, USA
Astronomy Tea Talk Cahill Center for Astronomy and Astrophysics, California Institute of Technology	Nov 2025 Pasadena, USA
Gemini North Observatory Talk Gemini Observatory, NSF NOIRLab	Nov 2025 Hilo, USA

Schools

18th Heidelberg Summer School	Sep 2023
Max Planck Institute for Astronomy, University of Heidelberg (<i>Participant</i>)	Heidelberg, Germany
Science Leadership Conference	Sep 2020
The Optical Society (Optica) (<i>Participant</i>)	Remote
Sixth Byurakan Summer School	Sep 2018
Byurakan Astrophysical Observatory (<i>Participant</i>)	Byurakan, Armenia
Wolfram Summer School	Aug 2017
Wolfram Research Company (<i>Participant</i>)	Yerevan, Armenia

Local Organizing Committee

VHE Phenomena around Supermassive Black Holes Workshop	Sep 2019
Russian-Armenian University, University of Heidelberg, Volkswagen Foundation	Yerevan, Armenia
IONS International Conference	Jun 2017
OSA YSU and NAS Chapter	Yerevan, Armenia

NON-ACADEMIC EXPERIENCE

Education Expert/Lead Physics Teacher	2020 — 2022
National Center of Educational Technologies	Yerevan, Armenia
- Mentored teachers in implementing remote teaching methods, developed quality assurance tools for educational programs. - Made 100+ videos based on a physics school program available on YouTube.	
Teacher Trainer/Mentor	2019 — 2022
State Pedagogical-Psychological Center	Yerevan, Armenia
Mentored educators in effective classroom management techniques and wrote a handbook on STEM teaching methods.	
Physics/Astrophysics Teacher	2018 — 2019
Shirakatsy Lyceum International Scientific-Educational Complex	Armenia
Led “Young Astrophysicist” student club and mentored students in scientific research projects.	
Robotics Workshop Leader	2017 — 2018
TUMO Center for Creative Technologies	Armenia
Taught robotics to children aged 12-18 and developed STEM curriculum.	

SKILLS

- Languages:** Armenian (Native), Russian (Fluent), English (Fluent), Spanish (Intermediate)
- Technical:** Python (matplotlib, pandas, astropy, scipy), Ruby, Wolfram Mathematica, LePhare, EsoReX, LaTeX, Git
- Soft Skills:** Scientific Communication, Teaching, Research Leadership, Project Management

AWARDS

Beca Adelina Gutiérrez	2024
Sociedad Chilena de Astronomía	Chile
Finalist of Falling Walls Science Engage	2020
International science communication competition	Global
Scholarship after Viktor Hambardzumyan	2020
Prestigious astronomy scholarship	Armenia
Scholarship after Gevorg and Gohar Vardanian	2014
Academic excellence scholarship	Armenia
Annual Presidential Award of the RA in IT Sphere	2014
Recognition for contributions to information technology	Armenia

REFERENCES

Dr. Chiara Mazzucchelli <i>Supervisor, Universidad Diego Portales, Santiago, Chile</i> chiara.mazzucchelli@mail.udp.cl	Dr. Roberto Assef <i>Co-Supervisor, Universidad Diego Portales, Santiago, Chile</i> roberto.assef@mail.udp.cl	Dr. Emanuele Farina <i>Collaborator, International Gemini Observatory, Hilo, USA</i> emanuele.farina@noirlab.edu
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Additional references available upon request